

Talk Outline

March 20, 2026

1. Motivation
2. Radiation interactions in silicon
3. Geant4 transport setup
4. From recoil events to defects
5. Defect-driven lifetime degradation
6. Diffusion length and leakage current trends
7. Device consequences
8. Limitations and future work

Cryogenic extension

Add a short ending section on how radiation-induced defect behavior changes at low temperature. Focus on migration freeze-out, trap emission freeze-out, and annealing freeze-out, and use simple MATLAB Arrhenius plots to show the trend from 300 K toward the cryogenic limit.